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Agentic Autoresearch for CT Reconstruction

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ABSTRACT

Background. Deep learning has produced a large and diverse body of CT reconstruction methods. Fair comparison across them remains labor-intensive and largely manual. Moreover, many benchmarks rely on idealized data. **Purpose.** We test two things. First, can a large language model (LLM) agent do the labor of reconstruction research — implement, tune, and benchmark many methods on its own? Second, does a ranking measured on ideal data predict how methods behave under realistic noise? **Methods.** We built an agentic loop. The agent edits a solver, runs a short job on a compute cluster, reads one frozen metric, and revises. The metric is a calibrated headroom score, $\mathrm{hr} = \max(0, 1 - \mathrm{RMSE}/\mathrm{RMSE}_{\mathrm{FBP}})$, computed inside the field of view. We grounded every method in the same differentiable fan-beam projector. We benchmarked 26 reconstruction methods on two problems: Mayo low-dose CT (noise-limited) and a 128-view sparse-view breast task from the noiseless DL-Sparse-View Challenge. We selected each method's best iteration on validation and report its score on the held-out test set (per-case mean $\pm$ standard deviation). To investigate the effect of noise in the noiseless breast CT challenge, we then re-evaluated every trained model on the same breast test cases with mild Poisson noise added to the inputs ($I_0 = 10^5$ photons), *without any retraining*. **Results.** Benchmarking gives a small tier of statistically indistinguishable top methods, not one winner. From the 26 methods the agent recombined a compact solver of $\sim$195 parameters that reaches the top accuracy tier at $\sim$2 % of its parameters. The best compact architecture is problem-dependent: a denoiser for noisy Mayo, a filtered data-consistency and learned primal-dual unroll for sparse-view breast. Mild input noise nearly inverts the breast ranking. The noiseless champion (supervised image denoiser, hr 0.89) collapses to last (hr 0.00). A learned primal-dual method rises from mid-pack to champion (hr 0.72 → 0.93). Physics-regularized and hand-crafted-smoothing methods rise; supervised image-domain denoisers collapse. **Conclusions.** An LLM agent can implement, tune, and benchmark reconstruction methods at scale under a fixed metric and compute budget. It did not invent new methods, and it required some human guidance to develop highly efficient parameter methods. More importantly, a leaderboard measured on ideal data does not predict robustness — it can invert under a small, realistic perturbation. We argue that benchmarks must include noise and other perturbations by default, and that agentic autoresearch makes such multi-condition benchmarking ready for routine use.

1 | Introduction

Deep learning has changed CT reconstruction. A decade of work has produced many learned reconstructors. Examples are unrolled iterative networks, learned primal-dual schemes, variational networks[1], image-domain and dual-domain denoisers, diffusion priors (generative models, unrelated to physical diffusion), and per-scene implicit representations.[1,2,3,4,5,6] Each promises a good image from fewer views or a lower dose than filtered back-projection (FBP). The best of them deliver. At first glance the research program looks simple: pick the strongest network, train it, reconstruct.

Yet, a purely data-driven reconstructor learns a map that is not tied to the physics. Such a map is free to *hallucinate*.[7] It can add anatomy that the measurements do not support.[8] This is not a small tuning issue. It is a structural risk. The field's durable

[1] A variational network (Hammernik et al.) is an unrolled network whose learned regularizer is a filter-bank "field of experts"; we use it here as the accuracy yardstick.

answer is to put the physics back in. We embed the **known operator** — the differentiable forward and back projector — into the learned pipeline.[9,10] In fact, this trade is quantified. Constraining a network with an exact operator provably cannot raise, and usually lowers, the maximum error bound. In practice it also lowers the number of parameters and the amount of training data needed.[9] A recent deep risk estimator establishes this benefit on theoretical grounds.[11]

There is a second problem, and it sits one level up. The body of methods is itself the product of slow, largely manual work. The reconstruction problem can be framed as a learned inverse problem,[6] but to compare or improve any method, a researcher must implement a solver, wire it to the data and geometry, launch a job, read a metric, form a hypothesis, change one thing, and repeat. This takes days per method. Recent "autoresearch" ideas imagine automating this loop — an automated research cycle in which the software itself proposes a change, runs the experiment, reads the result, and iterates.[12] Now, large language model (LLM) agents offer game-changing capacities. A large language model is a neural network trained to predict text. An *agent* wraps such a model in a loop so that it can act — read files, write code, and launch jobs — not just answer questions; a loop driven this way we call *agentic*. General-purpose coding agents already resolve real software-engineering tasks,[13,14,15] and physics-informed agentic development has begun in adjacent imaging fields.[16] *Can an LLM agent do reconstruction research?* Not just tune one number, but implement an unfamiliar method, diagnose why a reconstruction fails, edit its own solver code, and benchmark the result fairly — if we ground it in the same two things that would also guide a human: a differentiable projector and a fair, frozen metric.

We built such a loop and let an agent run it. With it, we benchmarked 26 methods across two CT problems. One is noise-limited (Mayo low-dose[17]); the other is incompleteness-limited (128-view sparse-view breast, from the noiseless DL-Sparse-View Challenge[1]). We then asked a second question that the ideal-data benchmark cannot answer on its own: does the ranking survive a little noise? To answer it, we added mild Poisson noise to the test inputs and re-scored every trained model without retraining.

This paper makes five contributions. (i) An agentic CT-reconstruction loop grounded in a differentiable projector and a frozen calibrated metric, with full provenance. (ii) An autonomous implementation and fair benchmark of 26 methods across two regimes. (iii) An evidence-derived compact solver, competitive with the top tier at ~1–2 % of its parameters, whose optimal form is problem-dependent. (iv) A robustness result — ideal-data rank does not predict, and can invert under, noise — and the practical lesson it carries for how we build benchmarks. (v) We release all 26 methods, our compact solver, and the agentic loop as open source (https://github.com/akmaier/Agent4CT).

2 | Materials and Methods

2.1 | The agentic loop

The agent is driven by a single large language model, Claude Opus 4.8 (Anthropic; model `claude-opus-4-8`, 1M-token context), throughout. It works in a fixed cycle. First it reads the previous result and names the failure. By *failure* we mean the specific way the current best reconstruction falls short of the frozen metric — a stalled or low headroom score, or a dominant error mode read off the comparison figure (residual streaks, over-smoothing, or noise amplification) — which the agent then targets. It changes one thing — a hyper-parameter, or the solver code itself — and states a hypothesis. A single short cluster job runs under a fixed compute budget. The agent reads the frozen metric and accepts or discards the change. The cycle then repeats. Every run is grounded in the same *differentiable* fan-beam projector (PYRO-NN[18]) — a forward/back-projector built so that gradients flow through it, so it can sit inside a trainable network and be optimized by back-propagation. Each run also writes an immutable record: the configuration, the reconstruction, the metric, and a comparison figure. We enforced a fixed wall-clock budget per iteration (20 min on breast; matched on Mayo) so that all methods are compared under equal compute. A human coordinator supervised the loop: setting targets, redirecting across problems, and auditing for padding and provenance gaps (Section 4). The loop is sketched in Figure 1.

2.2 | A single framework for 26 methods

We describe the forward model first. For the reconstruction, the scanner is assumed as a linear operator. The sinogram is the set of line integrals of the image:

$$\mathbf{g} = \mathbf{A}\,\mathbf{x} + \boldsymbol{\varepsilon}. \tag{1}$$

Here $\mathbf{x}$ is the image, $\mathbf{A}$ the discrete fan-beam projector, $\mathbf{g}$ the measured projections, and $\boldsymbol{\varepsilon}$ the noise. For the noiseless Sidky breast data $\boldsymbol{\varepsilon} = \mathbf{0}$. Reconstruction means recovering $\mathbf{x}$ from $\mathbf{g}$. With 128 views, $\mathbf{A}$ is under-determined.

Almost every method is one instance of the same regularized inversion. We trade fit to the measurements against a prior:

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x}} \ \tfrac{1}{2}\|\mathbf{A}\mathbf{x} - \mathbf{g}\|_2^2 + \lambda\,\mathcal{R}(\mathbf{x}). \tag{2}$$

A method is fixed by two choices. The first is the **data-consistency operator D**, which maps the measurement residual back to image space and so pulls the image toward agreement with the sinogram. The second is the **prior** $\mathcal{R}$ (also called the regularizer): it encodes what a plausible image looks like and sets the fit-versus-smoothness trade-off. Many solvers approximate Eq. (2) by an *unrolled* proximal-gradient scheme of K steps. Unrolling fixes the iteration to K steps and makes each step a trainable network layer, so the whole fixed-length scheme is trained end-to-end:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha_k\,\mathbf{D}\big(\mathbf{A}\mathbf{x}_k - \mathbf{g}\big) + \mathcal{R}_\theta(\mathbf{x}_k). \tag{3}$$

Each step is a proximal-gradient update: a gradient step toward data fit through $\mathbf{D}$, then a "proximal" step — the learned prior $\mathcal{R}_\theta$ — that cleans the image. The form of $\mathbf{D}$ matters. Here, the raw adjoint $\mathbf{D} = \mathbf{A}^\top$ back-projects the residual and re-injects sparse-view streaks, whereas the filtered map $\mathbf{D} = \mathrm{FBP}(\cdot)$ (a ramp/Hann-filtered back-projection) does not. This distinction is the key lever on the sparse-view problem.

Some methods add a learned dual in the measurement domain. This is the learned primal-dual (LPD) block.[2] It learns updates in both the image domain (the *primal*) and the measurement/sinogram domain (the *dual*). A sinogram-space memory $\mathbf{h}$ and the

TABLE 1 | Solver taxonomy (condensed). Each family is placed on Axis A (physics engagement, i.e. how the operator **A** is used at inference) and Axis B (prior source $\mathcal{R}$), with a source citation per family. The full per-solver configuration and parameter counts are in Supplement S1.

Family	Axis A	Axis B	Ref.
Classical iterative (TV, PWLS-TV)	in-loop	hand-crafted	[20,21]
Image-domain denoiser (bilateral, U-Net)	none	hand-cr. / sup.	[22,23]
Unrolled learned-iterative (ITNet, U-Swin)	in-loop	supervised	[4,5,24,25,26,27]
Variational network (Hammernik)	in-loop	supervised	[28,3]
Learned primal-dual (LPD)	in-loop	supervised	[2]
Dual-domain (DD sup., bilateral, N2I)	single / in-loop	sup. / self-sup.	[29,30]
Generative prior (fastdiff ×4)	in-loop	generative	[31,32,33,34]
Per-scene implicit (NAF, R^2-Gaussian)	per-scene	implicit	[35,36,37]
Foundation zero-shot (RAM)	single	foundation	[19]
Band decomposition (Wu-2015)	in-loop	hand-cr. / sup.	[21]
Recombination (ours)	**in-loop**	**hybrid**	—

image **x** update together:

$$\mathbf{h}_{k+1} = \Gamma_\phi(\mathbf{h}_k,\ \mathbf{A}\mathbf{x}_k - \mathbf{g}), \qquad \mathbf{x}_{k+1} = \Lambda_\theta(\mathbf{x}_k,\ \mathbf{D}(\mathbf{h}_{k+1})). \quad (4)$$

The convolutions $\Gamma_\phi, \Lambda_\theta$ are small and weight-tied. We zero-initialize their final layer, so the block is the identity at the start and cannot regress.

Table 1 places all 26 established methods on two axes. Axis A is physics engagement: how much **A** is used at inference (none, a single data-consistency step, an in-loop unrolled scheme, or a full per-scene fit). Axis B is the prior source, and its categories are worth naming plainly. *Hand-crafted* priors (total variation, bilateral) encode smoothness by hand. *Supervised* priors are trained on paired examples with a known answer (low-dose input, full-dose truth). *Self-supervised* priors train without a clean reference, using structure in the data itself; Noise2Inverse (N2I), for example, splits the measured projections into subsets so one subset's reconstruction predicts another's. *Generative* priors are diffusion models — networks that make realistic images by reversing a gradual noising process — used to "imagine" plausible detail. *Implicit/per-scene* methods fit a small network to each single scan that returns the attenuation at any queried coordinate, rather than storing a pixel grid; neural fields and Gaussian splatting (which models the volume as many small 3-D Gaussian blobs) are examples. A *foundation model* is a large network pre-trained once on generic data and applied here with no task-specific training ("zero-shot"); we include the Reconstruct Anything Model.[19]

This table also names the agent's job in the compact study: re-select the pair $(\mathbf{D}, \mathcal{R})$. Full per-solver configurations are in Supplement S1.

Search protocol. Each of the 26 established methods was tuned by 20 autoresearch iterations of the loop in Section 2.1. The 27th method — our compact solver — was searched for 40 iterations, with the explicit aim of an ultra-low-parameter recombination of the pair $(\mathbf{D}, \mathcal{R})$. During its first 20 iterations the agent worked alone and was free to recombine any component of the other 26 solvers. During the second 20, a human operator added mild suggestions about directions not yet tried. This was the only stage where the human shaped the search beyond setting targets and auditing.

2.3 | Datasets and geometry

Mayo low-dose CT. Real helical low-dose data from the 2016 AAPM Low-Dose CT Grand Challenge.[17] Rebinning helical to fan-beam via single-slice rebinning[38] allows direct use of the fan-beam projector. We built and validated this pipeline, and use a fixed train/val/test split (Supplement S2). This is one of a family of AAPM reconstruction grand challenges spanning spectral, truth-based, and metal-artifact tasks.[39,40,41]

Breast CT Challenge. 128-view 2-D fan-beam sparse-view data of synthetic breast phantoms.[1] This data is **noiseless by design**. The challenge asks whether deep learning can solve the sparse-view inverse problem, so its RMSE floor is exactly zero. We re-partitioned the public 4000-case train pool into train (3600), validation (200), and test (200), with a fixed seed and disjoint cases. To confirm the split, we verified disjointness by image hashing: no test image appears in train or validation (Supplement S3).

2.4 | The calibrated-headroom metric

We score how far a method closes the gap between the low-dose FBP and the truth. Let **M** be a field-of-view mask. The headroom is

$$\mathrm{hr} = \max\Big(0,\ 1 - \frac{\mathrm{RMSE}(\mathbf{M}\odot\hat{\mathbf{x}},\ \mathbf{M}\odot\mathbf{x})}{\mathrm{RMSE}(\mathbf{M}\odot\mathbf{x}_{\mathrm{FBP}},\ \mathbf{M}\odot\mathbf{x})}\Big). \quad (5)$$

Here **M** is the circular scan field-of-view mask, and $\mathbf{x}_{\mathrm{FBP}}$ is the filtered-back-projection baseline computed from the same input as $\hat{\mathbf{x}}$ — the clean sinogram on the noiseless board and the noisy sinogram on the noisy board. An hr of 0 means no better than that FBP baseline: the $\max(0,\cdot)$ floors any method that fails to beat it. An hr near 1 means near-perfect. We select each method's best iteration on validation, then report its scores once on the held-out test set as the per-case mean ± standard deviation. SSIM and PSNR use a batch-wide data range so that they are comparable across cases.

2.5 | The noise-robustness experiment

We ask whether the noiseless ranking survives a little input noise. For each method we take its noiseless best iteration. We then re-evaluate it on the **same** 200 breast test cases, but with photon noise added to the sinograms. For a line integral p and incident photon count I_0,

$$N \sim \mathrm{Poisson}(I_0\, e^{-p}), \qquad \hat{p} = -\log(\max(N, 1)/I_0). \quad (6)$$

We use $I_0 = 10^5$ photons. This is a rather high dose. The noise is mild — about 1–2 % at the thickest ray. We apply it to the test inputs only. Note that the ground truth stays clean. Supervised solvers **load their noiseless-trained weights and no additional training is applied**. Per-scene and classical solvers re-fit on the noisy input, which is their normal inference. The FBP baseline in Eq. (5) is recomputed from the same noisy data, so hr measures improvement over the noisy FBP. This is exactly the realistic model $\mathbf{Ax} + \varepsilon$ that the noiseless challenge does not study.

2.6 | Statistics

We begin with Mayo. There the five held-out patients give $n = 5$, so the paired t-test has low power; we therefore report both the 5 % and 1 % levels. For Breast CT, the 200 test cases are shared by all methods, so comparisons are paired. With $n = 200$ the paired t-test is very high-powered: tiny mean gaps reach $p < 0.01$. Hence, we report **effect size** — how large a difference is, independent of sample size — not just p. We use Cohen's $d_z = \bar{d}/s_d$, the mean of the paired differences over their standard deviation, and lead with d_z and the raw difference. The contrast between $n = 5$ and $n = 200$ is discussed in Section 4.

3 | Results

3.1 | Mayo low-dose CT: a tier, not a winner

On Mayo, the leaderboard does not yield a single winner. It yields a small top tier that is statistically indistinguishable. The champion is ITNet at hr 0.376 ± 0.089, with ITNet-v2 (0.374) and U-Swin (0.370) inside one standard error. A paired t-test cannot separate ITNet, its variants, and U-Swin at the 5 % level, and our compact recombination solver (Section 3.3) joins this tier at the 1 % level ($p = 0.02$–0.027). The reason is sample size: the Mayo test set has five patients, so $n = 5$, and at that size only large, consistent gaps clear significance — the paired t-test we use does separate the wider gaps below the tier, but the small, patient-to-patient-noisy differences among the top methods do not (a two-sided Wilcoxon signed-rank test cannot reach $p < 0.05$ at all at $n = 5$). Hence the top of the board is best read as a tie, not a strict order. Note that the absolute image metrics agree with this reading. The top six solvers all exceed SSIM 0.97 and sit at the RMSE floor of 5×10^{-4}, so SSIM saturates exactly where the calibrated headroom still separates methods. Below the tier the board spreads cleanly: the variational network reaches hr 0.159, classical total variation 0.096, and the PWLS-TV and bilateral baselines fall further. Several families — diffusion priors, per-scene implicit methods, the zero-shot foundation model, and the primal-dual unroll as configured for the Mayo geometry — do not clear the low-dose FBP baseline and sit at hr 0, even though their SSIM stays high (0.82–0.95); SSIM rewards the smoothness that hr, measured against FBP, does not credit. The full board, the per-patient headroom, and the pairwise significance matrix are in Table 2 and Supplement S4. One practical result deserves emphasis, and it concerns effort. Every line of code in this study — the geometry pipeline, all solvers, and the benchmark harness — was written by the agent; the human did not hand-code, but instructed. What varied, by a wide margin, was how much instruction each part needed. The helical-to-fan *geometry pipeline* was by far the hardest: bringing the differentiable rebinning and its calibration to a physically correct FBP baseline took many supervised iterations and was the dominant human cost of the study (Supplement S2). *Implementing and benchmarking the 26 solvers* needed far less: the agent onboarded 13 of them across two days, and the three-week benchmark campaign was overwhelmingly automated — most of its commits change only results, not code — while the agent ran ∼1,100 iterations for ∼134 GPU-hours of compute, largely in parallel. Costing each source-changing commit at an estimated 7.5 minutes (Supplement S2), the human's hands-on share of the solver and benchmarking work was on the order of hours. The asymmetry is the lesson: instructing an agent is far cheaper than coding by hand — though not uniformly, since the hard, physics-heavy pipeline still demanded the most human supervision while 25 methods layered on top cost hours.

3.2 | Noiseless Breast CT Challenge: the numbers are not a leak

On the noiseless Breast CT Challenge the headroom values are high. The champion is the supervised dual-domain denoiser at hr 0.89, with the ITNet family (0.87–0.89) and U-Swin (0.86) just behind; the top five solvers all exceed 0.85 (Table 3, left board). This is a property of the data, not a flaw and not a leak. The Breast CT Challenge data is noiseless by design, so its RMSE floor is exactly zero and near-perfect recovery is possible in principle. Because such numbers invite suspicion, we ruled out train/test leakage two ways. By construction, the public pool is partitioned by index under a fixed seed, so no case can appear in more than one split; and by mechanism, the training loader only ever reads the train split, with the test redirect firing solely at evaluation, so no model sees a test image while training. Hence the high numbers reflect an easy, well-posed, incompleteness-limited problem, not memorization. This also means the breast and Mayo hr scales are not comparable. Breast is limited by missing views (128 of ∼1000), Mayo by real quantum noise, so a breast hr of 0.89 and a Mayo hr of 0.38 describe different bottlenecks, not different amounts of the same thing (Supplement S3).

3.3 | A compact, problem-dependent solver

Beyond ranking the 26 established methods, the agent recombined their parts into a compact solver — and the best recombination is problem-dependent. On Mayo, a noise-limited problem, the compact optimum is a small multi-scale bilateral denoiser of about 970 parameters (hr 0.324) that joins the top tier at the 1 % level; where the bottleneck is noise, a tiny denoiser suffices. On breast, a sparse-view problem, the optimum is a different object entirely. It is a filtered data-consistency step ($\mathbf{D}$ = FBP applied to the residual $\mathbf{A}\hat{\mathbf{x}} - \mathbf{g}$), a learned cross-step combination in the primal-dual style (Eq. 4), and a small multi-scale bilateral filter on the output. The climb to it was made of single-knob iterations, each earned against the frozen metric. A pure image-domain denoiser, however configured, capped at hr ≈ 0.26. Replacing the adjoint by the *filtered* gradient broke that ceiling with only 13 parameters (hr 0.43). A learned cross-step combination added headroom (183 parameters, hr 0.50). A three-scale bilateral output filter completed the design (195 parameters). The final breast solver reaches hr 0.6212 ± 0.0076 — the variational-network accuracy tier at ∼2 % of that network's parameters, and with the tightest per-case standard deviation of any solver on the board. The two compact solutions do not transfer. On a sparse-view problem data-consistency is the differentiator; on a noise-limited problem it is not. This problem-dependence is the practical face of the known-operator principle: embedding the exact forward operator cannot raise, and usually lowers, the maximum error bound, and it lets a few hundred parameters do the work of millions.[9] In fact, the agent re-derived each optimum from its own runs; when steered toward the Mayo answer on breast it did not transfer, because the evidence on breast pointed elsewhere (Supplement S5). Figure 2 places both compact solvers against parameter count.

TABLE 2 | Mayo low-dose CT test board (all 26 solvers; $n = 5$ test patients). hr is the calibrated headroom (Eq. 5) and the ranking metric; SSIM and RMSE use a batch-wide data range (RMSE in line-integral units). Params are trainable-parameter counts; entries suffixed M are in millions, an unsuffixed number is a raw count. "–" marks a solver below the ranking floor (hr $\leq$ 0). Solver keys abbreviated: dd = dual-domain, px = pixel, wav = wavelet, dc = data-consistency step, con./unc. = (un)constrained, sup = supervised.

Rank	Solver	hr	SSIM	RMSE	Params
1	itnet	0.3756	0.9790	0.0005	0.233 M
2	itnet-v2	0.3735	0.9784	0.0005	0.233 M
3	uswin	0.3700	0.9770	0.0005	3.954 M
4	dd-supervised	0.3607	0.9758	0.0005	0.466 M
5	param-efficient	0.3241	0.9727	0.0005	969
6	itnet-v3	0.3066	0.9707	0.0006	3.699 M
7	hammernik-vn	0.1591	0.9520	0.0007	0.005 M
8	tv-iterative	0.0959	0.9642	0.0007	0
9	hammernik-2017	0.0843	0.9358	0.0007	0.012 M
10	manhart-pwls-tv	0.0624	0.9649	0.0007	0
11	wu-2015-trainable	0.0207	0.8923	0.0008	8
12	dd-n2i	0.0176	0.9590	0.0008	0.466 M
13	manduca-bilateral	0.0159	0.9561	0.0009	7
14	dd-bilateral-sup	0.0040	0.9541	0.0009	12
–	tv-iterative-sup	0.0109	0.9366	0.0008	2
–	dd-bilateral-n2i	0.0000	0.9587	0.0010	12
–	fastdiff-flow-px-unc	0.0000	0.9523	0.0012	3.823 M
–	fastdiff-wdm-wav-con	0.0000	0.9501	0.0013	3.825 M
–	ram	0.0000	0.9446	0.0012	35.619 M
–	learned-primal-dual	0.0000	0.9415	0.0011	0.155 M
–	fastdiff-wdm-wav-unc	0.0000	0.9404	0.0013	3.825 M
–	fastdiff-flow-px-con	0.0000	0.9300	0.0013	3.823 M
–	diff-recon-dc-unc	0.0000	0.9084	0.0014	8.594 M
–	diff-recon-dc-con	0.0000	0.8766	0.0016	8.594 M
–	r2gaussian	0.0000	0.8234	0.0043	0.003 M
–	naf	0.0000	0.8172	0.0028	33.558 M

3.4 | A little noise inverts the ranking

We now perturb the inputs and re-score, with no retraining. We add mild Poisson noise to the 200 breast test sinograms at $I_0 = 10^5$ photons — about 1–2 % at the thickest ray — keep the ground truth clean, reload each model's noiseless-trained weights, and recompute the FBP baseline from the noisy data so that hr measures improvement over the *noisy* FBP. Nothing else changes. The clean ranking does not predict the noisy one. Across the solvers ranked on clean data, the two orderings are almost uncorrelated (Spearman $\rho \approx 0.14$; the clean ranking explains only ~2 % of the variance in noisy rank). The noiseless champion drops to last, and of the five best solvers on clean data only ITNet-v3 survives in the noisy top five; the rest are displaced by physics-regularized and hand-crafted-smoothing methods. Table 3 shows the two boards side by side. The headline entries:

- **learned-primal-dual**[2] rises from hr 0.72 (rank 6) to **0.93** (rank 1), with noisy SSIM 0.99. It keeps the forward operator in the loop, so it re-solves the noisy measurements rather than denoising a fixed image.
- **manduca-bilateral**, a 22-parameter hand-crafted smoothing prior,[22] rises from 0.28 to **0.84** (rank 2). It has nothing to overfit, so smoothing is noise-robust by construction. The self-supervised dual-domain bilateral (dd-bilateral-n2i)[30] rises even more sharply, from an essentially failed 0.0001 on clean data to **0.79** (rank 3).
- **dual-domain-supervised**, the noiseless champion (0.89, rank 1),[29] **collapses to hr 0.00** and out of the ranking, with SSIM 0.35. It is a pure supervised image-domain denoiser trained only on clean FBP, so a distribution it never saw breaks it.
- The supervised unrolled family (ITNet v1/v2/v3, U-Swin) drops from 0.86–0.89 into the middle of the board (0.55–0.70). Our compact solver holds relatively well (0.62 → 0.52, rank 13).
- Most strikingly, the unconstrained wavelet diffusion prior (fastdiff-wdm-wav-unc) — a deliberate negative control that *failed* on clean data (hr 0, below the FBP floor) — is noise-robust and ranks 8th (hr 0.64, SSIM 0.95). Its generative prior never fit the clean forward model, so it degrades gracefully; the zero-shot foundation model (RAM)[19] rises for the same reason (rank 7).
- Per-scene neural-field and Gaussian-splatting methods (NAF, R^2-Gaussian)[35,36] do not complete under the 20-minute budget on either board and are reported as DNF.

The absolute SSIM and PSNR columns confirm the reordering is genuine, not an artifact of the changing FBP baseline: learned-primal-dual reaches SSIM 0.99 on the noisy data while dual-domain-supervised falls to 0.35 (Supplement S6). Because every method sees the identical noisy input and shares the same noisy FBP baseline, the comparison is fair within the noisy board. Figure 3 shows the full reordering as a paired rank plot, and Figure 4 shows its pixel-level face for the two methods at opposite ends.

3.5 | The framework explains the reversal

The reversal is not a surprise once we read it off the two axes of Table 1. Brittleness concentrates in one corner: supervised priors with weak physics engagement — image-domain maps trained on clean FBP. Dual-domain-supervised is the extreme case. It applies a learned image-to-image map with no forward operator at inference, so a small input shift it never saw during training moves it off its learned manifold and the reconstruction collapses. The constrained diffusion variants, which also lean on a clean-data fit, fall the same way. Robustness concentrates where the prior is not tuned to the clean distribution, and two disjoint mechanisms reach it. The first is in-loop physics: learned-primal-dual and the variational networks keep the forward operator **A** in the loop, so they re-solve the noisy measurements instead of denoising a fixed image, and their ranking rises or holds. The second is a prior that never fit the clean data at all: hand-crafted smoothing (bilateral, total variation) has nothing to overfit, and even the generative and foundation priors that failed on clean data degrade gracefully precisely because they never matched the clean forward model. As such, the framework turns the reversal from a curiosity into a prediction. A method's position on the physics-engagement and prior-source axes tells us, before any noise is

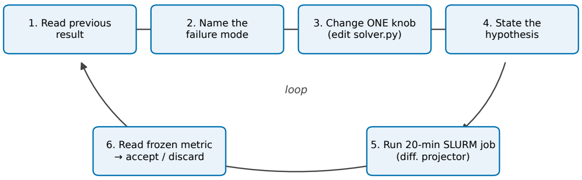

FIGURE 1 | The agentic autoresearch loop and its provenance. The agent reads the previous result, names the failure, changes one thing, states a hypothesis, runs one short cluster job grounded in the same differentiable fan-beam projector, reads the frozen calibrated-headroom metric, and accepts or discards the change. Every iteration writes an immutable record (configuration, reconstruction, metric, comparison figure).

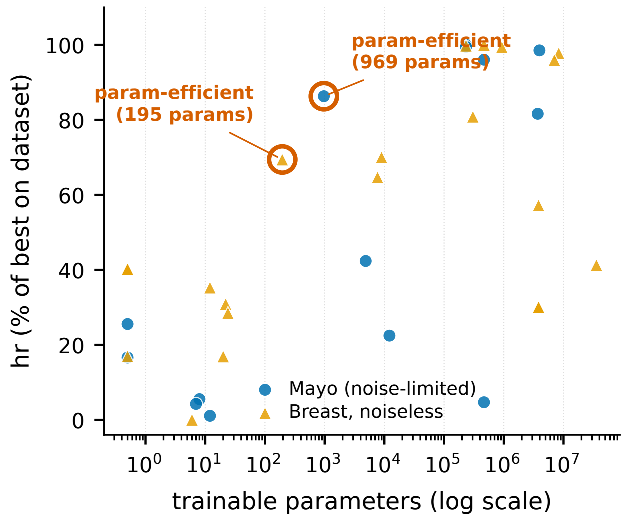

FIGURE 2 | Accuracy versus parameter count for both datasets (Mayo and breast noiseless). To overlay the two different hr scales, each solver's hr is normalized to the percent of the best hr reached on its own board. The agent's compact recombination solver is ringed: 969 parameters on Mayo, 195 on breast. It reaches the top accuracy tier at ~1–2 % of the parameters of the variational network.

added, whether it is likely to be brittle or robust. The lesson for benchmarking follows directly: a leaderboard measured on a single, idealized distribution rewards exactly the methods most likely to fail under a small, realistic perturbation.

4 | Discussion

4.1 | What the agent did well, and what it did not

We give a scorecard. It is the core of this Discussion.

The agent is strong in four ways. First, it is fast from paper to code: it turns a method description into a working solver quickly. The version-control history quantifies this. The agent onboarded 13 of the 26 established solvers across only two days (15–16 May 2026) and the rest in small batches; from a published method to a running, benchmarked solver was a fraction of a day each (per-solver dates in Supplement S2). This is the main speed-up, and

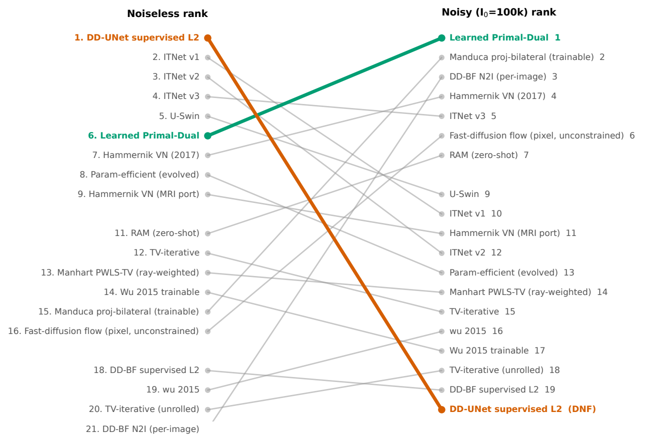

FIGURE 3 | The reversal. Paired rank plot from the noiseless breast board to the noisy breast board ($I_0 = 10^5$, same trained models, same 200 test cases, no retraining), with an arrow per method. The supervised image-domain champion collapses from rank 1 to last; the learned primal-dual method rises from mid-pack to champion; hand-crafted smoothing and in-loop physics priors rise. A small, realistic noise perturbation reorders almost the whole leaderboard.

it is why 26 methods across two datasets was feasible. In fact, it is also strong at hyper-parameter optimization. It respects a fixed compute budget, which is hard for humans and easy for the agent. And it scales evaluation: running many benchmarks in parallel is its strongest practical use.

The agent is weak in several ways. It does not invent new methods. Rather, it implements, tunes, and recombines known ones, and proposed no new reconstruction principle. Yet, it had to be *forced* to mix and match. The idea of combining proven pieces into one compact solver was a human idea, on both datasets. The agent executed it well but did not originate it. Left alone, it converges early and stops exploring. The compact-solver search shows this directly: the agent's own first 20 iterations plateaued at hr ≈ 0.50, and the decisive gains — the break to hr 0.62 — came only after a human began suggesting untried directions from iteration 20 on (Supplement S5). Its CT-image vision is unreliable: it keeps missing very clear artifacts in slices and sinograms, so numbers had to be the source of truth. Long tasks must be decomposed into sub-tasks; otherwise even a one-million-token context is used up quickly. And it overfits to the task and metric. Of course, this last point is probably not only an agent problem — human researchers overfit to benchmarks too.

The division of labor is clear. On one side, the agent is a tireless, budget-respecting, self-modifying executor. On the other, the human remains the strategist and auditor — forcing breadth, supplying the recombination idea, redirecting across problems, decomposing the work, and checking for padding and provenance. That division is the answer to the title question.

4.2 | Two scientific findings

First, the best compact architecture is problem-dependent. The Mayo solution is a denoiser, whereas the breast solution is a filtered-data-consistency and primal-dual unroll. Again, the agent re-derived each; it did not transfer one to the other. This supports the known-operator view: the right inductive bias —

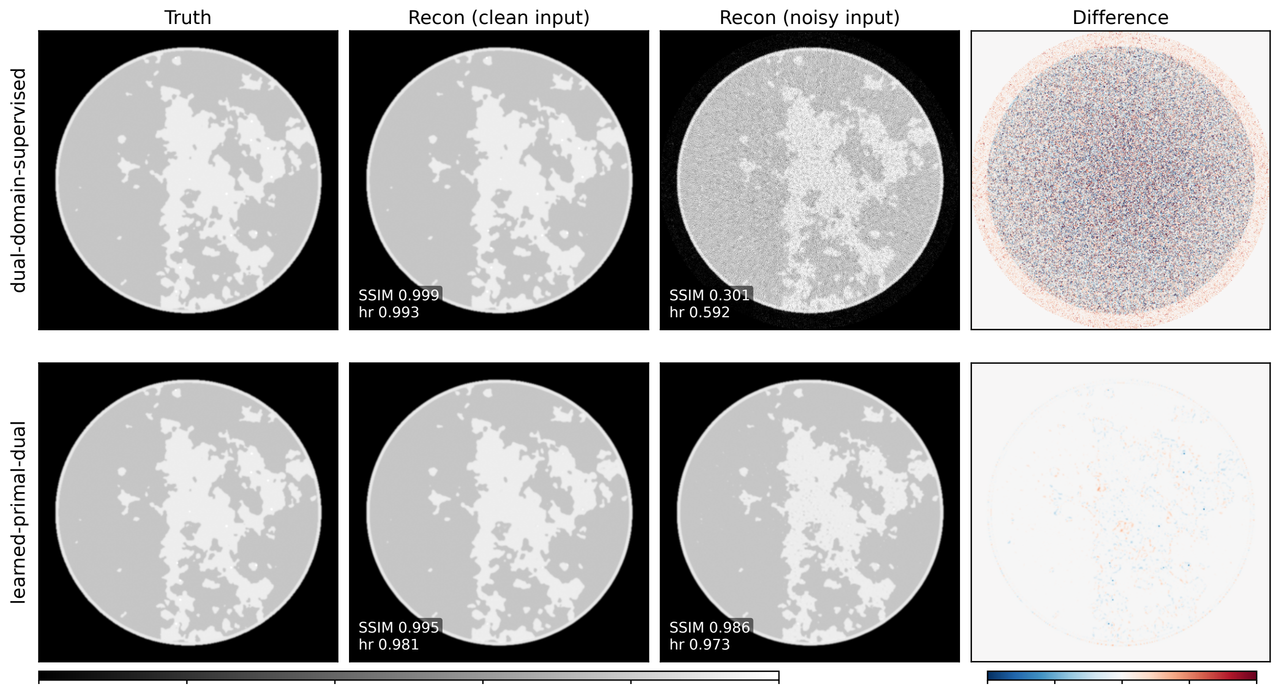

FIGURE 4 | Example reconstructions behind the reversal, for the two methods at opposite ends of it. Rows: the noiseless champion *dual-domain-supervised* (a supervised image-domain denoiser) versus *learned-primal-dual* (in-loop physics). Columns: ground truth, reconstruction from the clean input, reconstruction from the noisy input ($I_0 = 10^5$, same trained model, no retraining), and the difference to truth. Per-case SSIM and hr are annotated on each reconstruction; these are single-case values and differ from the 200-case board means in Table 3. On clean inputs both are near-perfect. Under mild input noise the supervised denoiser collapses (single-case SSIM 0.999 → 0.301), spraying structured error across the field of view, while the physics-in-the-loop primal-dual method holds (SSIM 0.995 → 0.986). This is the pixel-level face of the ranking reshuffle in Figure 3 and Table 3.

the assumptions an architecture builds in before it sees any data — depends on whether the problem is noise-limited or incompleteness-limited.

Second, a little noise broke the whole breast leaderboard. The methods that won on clean data were the least robust. This is a caution for the field. Significance also depends on sample size: at $n = 5$ (Mayo) the top methods tie; at $n = 200$ (breast) everything separates. As such, p-values are not comparable across datasets, and effect size should lead.

4.3 | Limitations

We used a single agent and a single frozen metric, under a fixed short per-iteration compute budget. The geometry/data-engineering bottleneck remains; the agent did not remove it. The agent-written solvers reproduce each method's design but may differ in detail from its official implementation. Our breast split is an internal partition of the public pool, not the withheld official challenge test set, and the Mayo comparison rests on only five test patients ($n = 5$), which limits statistical power. Some compact solutions are seed-fragile, and two per-scene methods did not complete under the budget. The noise study is deliberately narrow: we do not retrain or augment the supervised methods with matched noise — a condition that would likely restore much of their robustness, and whose effect on clean-trained networks is already documented [7] — and we use a single dose level ($I_0 = 10^5$) rather than a full dose sweep, which we leave to future work. Finally, we report no clinical or diagnostic endpoint.

5 | Conclusions

An LLM agent can implement, tune, and benchmark 26 CT reconstruction methods under one frozen metric and a fixed compute budget. The outcome of such a benchmark is a small tier of indistinguishable top methods, not one winner. Still, the agent does the labor; it does not replace the human strategist.

The more important message is about benchmarks. We used the agent to scale evaluation, and that scale exposed a problem. Our ideal-data leaderboard rewarded brittle methods. A small, realistic noise perturbation reordered almost the whole board. The methods that won on clean data collapsed. Hence, we argue that benchmarks must change. They should include noise, dose variation, and other realistic perturbations by default, so that leaderboards reward robust methods rather than methods that overfit ideal data. Agentic autoresearch makes such multi-condition benchmarking cheap enough to be routine. That is its real payoff for the field.

Acknowledgments

We thank the organizers of the AAPM Low-Dose CT Grand Challenge (Mayo Clinic), Emil Sidky, and the organizers of the Breast CT Challenge for making their data available.

Conflicts of Interest

The authors declare no relevant conflicts of interest.

TABLE 3 | The two breast boards side by side, for the same trained models on the same 200 test cases with *no retraining*. Left: noiseless board. Right: noisy board ($I_0 = 10^5$). The ranking nearly inverts. hr is the calibrated headroom (Eq. 5) and the ranking metric; SSIM and RMSE use a batch-wide data range (RMSE in line-integral units). "–" marks a solver that did not clear the ranking floor (hr $\leq$ 0) or did not finish (—). Params (M suffix = millions, otherwise a raw count) and solver keys as in Table 2.

Noiseless board

Rank	Solver	hr	SSIM	RMSE	Params
1	dd-supervised	0.8948	0.9992	0.0005	0.466 M
2	itnet	0.8926	0.9991	0.0006	0.233 M
3	itnet-v2	0.8893	0.9991	0.0006	0.928 M
4	itnet-v3	0.8749	0.9989	0.0007	8.318 M
5	uswin	0.8586	0.9986	0.0007	7.025 M
6	learned-primal-dual	0.7233	0.9962	0.0014	0.304 M
7	hammernik-2017	0.6265	0.9902	0.0019	0.009 M
8	param-efficient	0.6212	0.9912	0.0020	195
9	hammernik-vn	0.5787	0.9865	0.0022	0.008 M
10	fastdiff-flow-px-con	0.5119	0.9792	0.0025	3.823 M
11	ram-zeroshot	0.3693	0.9787	0.0033	35.619 M
12	tv-iterative	0.3601	0.9918	0.0033	0
13	manhart-pwls-tv	0.3601	0.9918	0.0033	0
14	wu-2015-trainable	0.3156	0.9537	0.0035	12
15	manduca-bilateral	0.2763	0.9830	0.0037	22
16	fastdiff-flow-px-unc	0.2693	0.9776	0.0038	3.823 M
17	fastdiff-wdm-wav-con	0.2693	0.9340	0.0038	3.825 M
18	dd-bilateral-sup	0.2546	0.9848	0.0039	24
19	wu-2015	0.1517	0.9611	0.0044	0
20	tv-iterative-sup	0.1513	0.9613	0.0044	20
21	dd-bilateral-n2i	0.0001	0.9583	0.0061	6
–	dd-n2i	0.0001	0.9461	0.0062	0.466 M
–	fastdiff-wdm-wav-unc	0.0000	0.9448	0.0112	3.825 M
–	naf	—	—	—	33.563 M
–	r2gaussian	—	—	—	0.006 M

Noisy board ($I_0 = 10^5$)

Rank	Solver	hr	SSIM	RMSE	Params
1	learned-primal-dual	0.9349	0.9905	0.0020	0.304 M
2	manduca-bilateral	0.8420	0.9537	0.0049	22
3	dd-bilateral-n2i	0.7927	0.9532	0.0064	6
4	hammernik-2017	0.7016	0.7178	0.0092	0.009 M
5	itnet-v3	0.7007	0.7564	0.0092	8.318 M
6	fastdiff-flow-px-unc	0.6620	0.7058	0.0104	3.823 M
7	ram-zeroshot	0.6382	0.6578	0.0111	35.619 M
8	fastdiff-wdm-wav-unc	0.6350	0.9450	0.0112	3.825 M
9	uswin	0.5915	0.6408	0.0125	7.025 M
10	itnet	0.5784	0.6295	0.0129	0.233 M
11	hammernik-vn	0.5529	0.5962	0.0137	0.008 M
12	itnet-v2	0.5480	0.6201	0.0139	0.928 M
13	param-efficient	0.5153	0.5771	0.0149	195
14	manhart-pwls-tv	0.5068	0.6164	0.0151	0
15	tv-iterative	0.5068	0.6164	0.0151	0
16	wu-2015	0.3700	0.5244	0.0193	0
17	wu-2015-trainable	0.2591	0.4400	0.0227	12
18	tv-iterative-sup	0.1071	0.4567	0.0274	20
19	dd-bilateral-sup	0.1003	0.4524	0.0276	24
–	dd-supervised	0.0000	0.3483	0.0338	0.466 M
–	fastdiff-wdm-wav-con	0.0000	0.3104	0.0477	3.825 M
–	fastdiff-flow-px-con	0.0000	0.2769	0.0583	3.823 M
–	dd-n2i	—	—	—	0.117 M
–	naf	—	—	—	33.563 M
–	r2gaussian	—	—	—	0.006 M

Data and Code Availability

The code — all 26 solver implementations, the compact recombination, the helical-to-fan rebinning pipeline, and the evaluation scripts — is open source at https://github.com/akmaier/Agent4CT, with the leaderboards and per-iteration provenance on the live dashboard at https://akmaier.github.io/Agent4CT. The challenge datasets are available from their original providers under their data-use terms and are not redistributed here: the Mayo low-dose CT data are the de-identified public AAPM Low-Dose CT Grand Challenge / TCIA collection, and the Breast CT Challenge data are synthetic phantoms. No new patient data were collected and institutional review board approval was not required; we comply with all challenge data-use agreements.

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